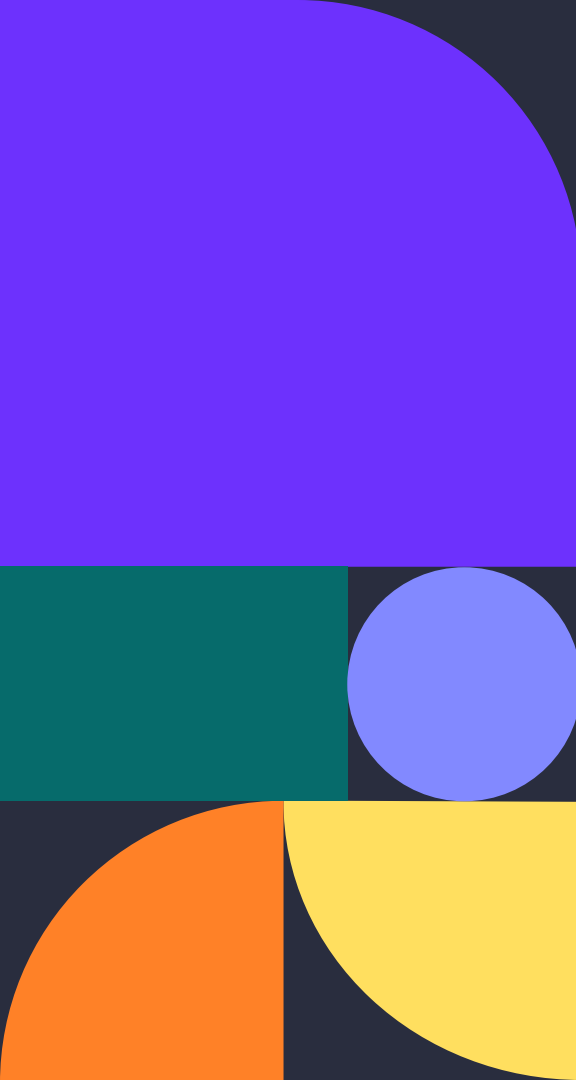




Transaction Anomaly Detection Pipeline

Arya K, Kiara J, Watson Blair

Agenda

01. **Introduction**

03. **About the Data**

05. **Continuous
Monitoring Pipeline**

02. **Related Topics**

04. **Core Anomaly
Detection**

06. **Results and
Conclusions**

Introduction:

Financial Fraud and Anomaly Detection

Wire Fraud and other Financial Reporting crimes rely on masking actual figures with a synthetic one to obscure a number of crimes.

With a throughput of hundreds of millions of transactions per year, fraud monitoring methodologies must be performant and unintrusive.

We propose a transaction monitoring pipeline that relies on progressive investigation and thresholding methodologies to isolate transactions with a high likelihood of fraud, without impeding authentic transactions.



Related Topics: Financial Domain



Benford's Law



In sets that obey the law, the number 1 appears as the leading significant digit about 30% of the time, while 9 appears as the leading significant digit less than 5% of the time.

Divergence from Benford's Law has been admitted in criminal cases at the federal, state and local levels as indication of financial crime.



Financial Dark Pools



Private trading forums, that allow institutional investors to trade large blocks of securities anonymously.

These transactions are inherently difficult to police and regulate.

Related Topics: Clustering



Gaussian Mixture Model



A mathematical model that represents non-normal probability distributions as the result of combined constituent normal distributions.

GMM Clustering is an Expectation Maximization algorithm that groups points based on the probability that they belong to a given cluster.



Isolation Forest



Selects the proportion of the population that is most unlike the rest of the population.

A clustering algorithm that groups points by the number of binary bisections it takes to isolate a given point in euclidean space.



Progressive Thresholding



A strategy for iteratively tuning thresholds in response to observations in the data.

In our pipeline we use progressive thresholding to drive the selection thresholds for both Gaussian Mixture Model(GMM) and Isolation Forest(IF).

About the Data:

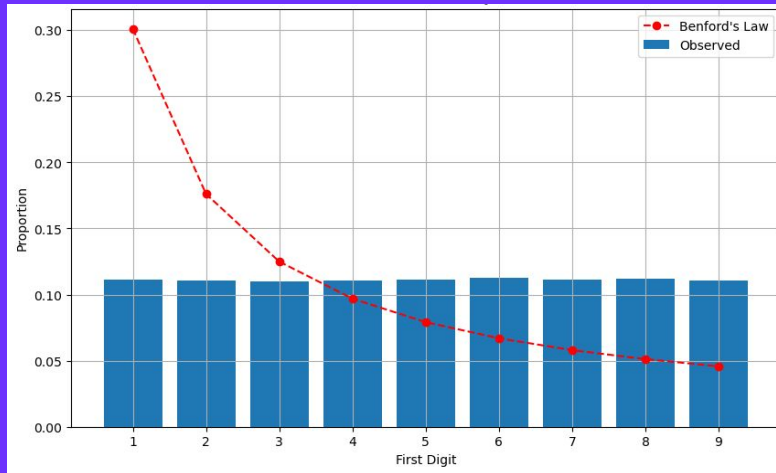
Financial Fraud and Anomaly Detection

The dataset covers anonymous transactions during 2023 between 15 unique accounts and 10 unique merchants across 5 locations using 3 transaction types. All merchants and accounts participate in all locations and transaction types. All accounts engage in transactions with all merchants.

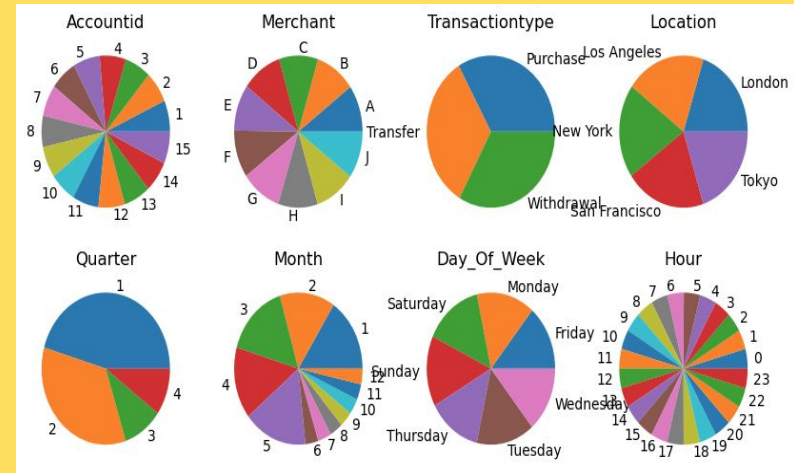
We further identified the transactions as Dark Pool Activity due to their anonymized nature. This characterization also makes the transaction types make more sense.

Data Statistical Features

Benford's Law



Total Amount



Anomaly Detection Methodologies



Clustered Anomalies

We use the union of GMM and IF clustering algorithms to select the points most probable to be anomalous



Conformity to Benford's Law

Statistical analysis of the dataset to ensure it conforms to the statistical profile of non-fraudulent data.



Divergence from historical trends

Statistical analysis of the dataset to ensure it conforms to the statistical profile of non-fraudulent data.





Continuous Monitoring Pipeline

01.

Examine transactions statistical parameters

02.

If anomalous, identify 5% of the dataset as anomalous.

03.

Reexamine statistical properties of remaining transactions

04.

If still anomalous, increase threshold by 5% and repeat process.

GMM

A probabilistic model that assumes all data points are generated from a mixture of a finite number of Gaussian distributions.

Overview

Optimizing Model

Threshold Logic

GMM Results

Soft Clustering: Assigns a probability of membership to each cluster.

Why GMM? Financial data is multimodal

Anomaly Metric:

Log-Likelihood. Lower likelihood indicates the transaction does not fit any learned normal pattern.

We used Information Criteria to balance goodness-of-fit against model complexity.

BIC (Bayesian Information Criterion)
AIC (Akaike Information Criterion)
Optimal Components $k = 10$

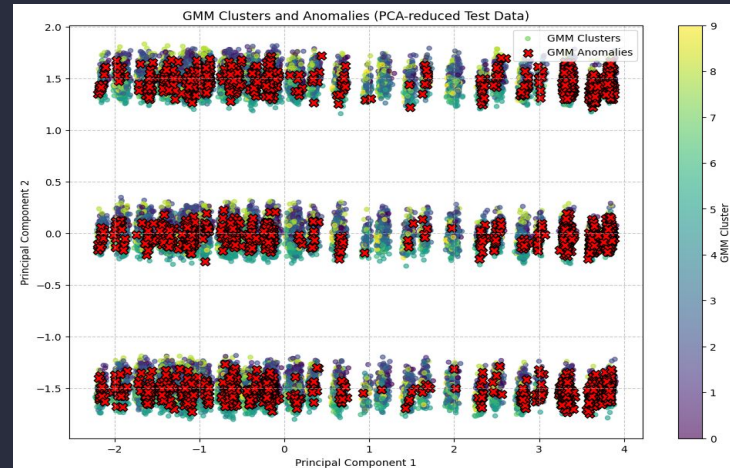
How do we define "Fraud"?

We trained the GMM on the training set and calculated the Log-Likelihood scores for all samples.

The Threshold: Set at the 5th percentile of training scores.

The Value: Any transaction with a log-likelihood score below ~ 23.64 is flagged as anomalous.

Anomalies Detected: 2,213
Anomaly Rate: 5.1%



Isolation Forest Algorithm

The IF algorithm was trained using the following parameters :

n_estimators: 200

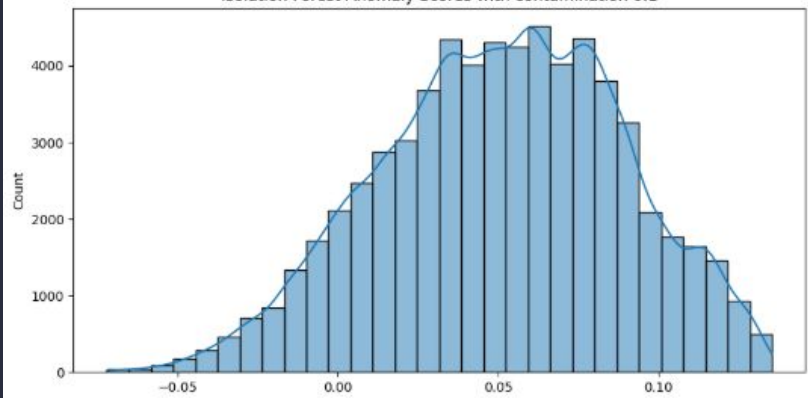
max_samples: 'auto'

contamination: tested with multiple values [0.1, 0.08, 0.05, 0.02]

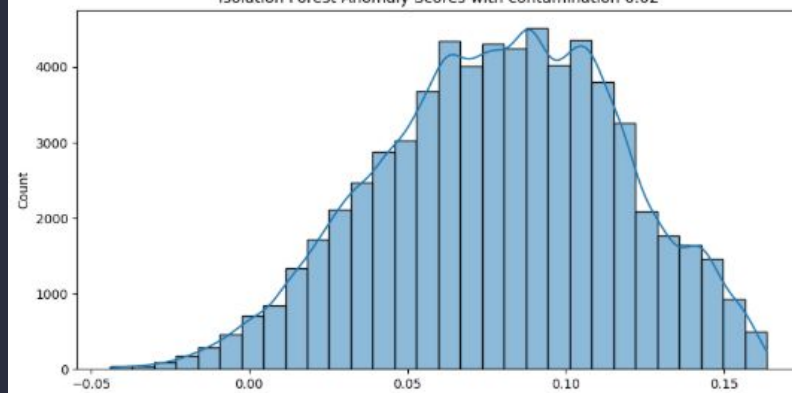
max_features: default

random_state: 42

Isolation Forest Anomaly Scores with contamination 0.1



Isolation Forest Anomaly Scores with contamination 0.02



IF Results

For contamination = 0.10, it flags 6504 anomalies

For contamination = 0.02, it flags only 1283 anomalies.

Higher contamination makes the model more aggressive, labeling more points as anomalies.
Lower contamination makes the model more conservative.

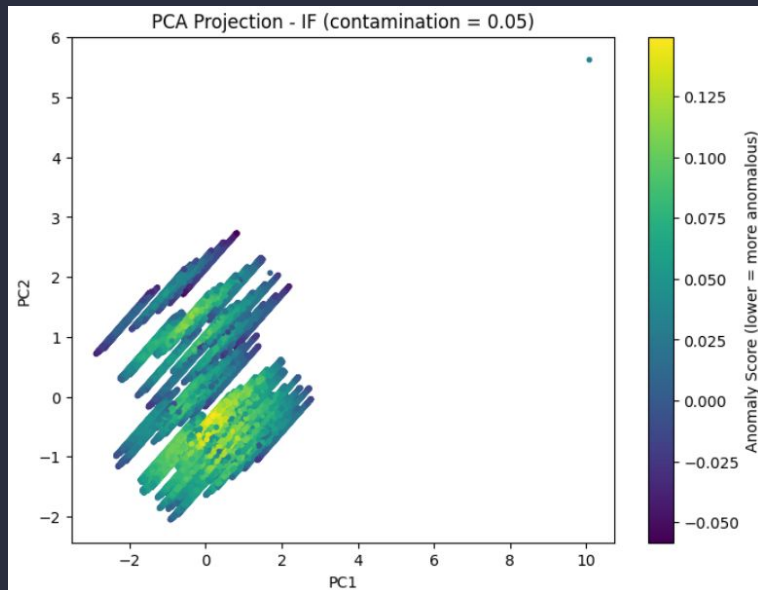
Most data points cluster tightly in a dense region, showing consistent normal behavior.

Only a few observations lie far from the main cluster.

These isolated points have much lower anomaly scores.

Results:

	contamination	Anomalies	Mean	Min	Max
0	0.10	6504	0.051095	-0.072110	0.135475
1	0.08	5197	0.055587	-0.067618	0.139966
2	0.02	1283	0.079215	-0.043990	0.163594
3	0.05	3197	0.064953	-0.058252	0.149332



Thank you